ZHAW School of Engineering
Institute of Mechatronic Systems

Betaflight – Offline Controller Tuning

Data-Driven Tuning of Rate and Altitude Controllers for FPV Drones

Bachelor Thesis

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Declaration of Originality

We hereby declare that this thesis is our own work and has been composed solely by the undersigned. We have not used any sources or aids other than those indicated. All passages taken verbatim or in substance from published or unpublished works of others have been clearly marked as such. This thesis has not been submitted, in whole or in part, for any other degree or examination at this or any other institution.

Winterthur, February 17, 2026

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Zusammenfassung

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Abstract

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Preface

This thesis was carried out during the spring semester 2026 at the Institute of Mechatronic Systems (IMS) at the ZHAW School of Engineering. It continues the work started in the preceding project thesis conducted in the fall semester 2025.

We would like to thank our supervisor, Michael Peter, for...

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Declaration of Generative AI Use

The following generative AI systems were used during the preparation of this thesis. Their use was limited to the functions described below. All generated content was critically reviewed and verified by the authors.

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